Problem A. Digit Sums

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Let $S(n)$ denote the sum of the digits in the decimal notation of n. For example, $S(101) = 1 + 0 + 1 = 2$.

Given an integer N, determine if $S(N)$ divides N.

Constraints

- $1 \leq N \leq 10^9$

Input

Input is given from Standard Input in the following format:

```
N
```

Output

If $S(N)$ divides N, print `Yes`; if it does not, print `No`.

Sample 1

Input	Output
12	Yes

In this input, $N = 12$. As $S(12) = 1 + 2 = 3$, $S(N)$ divides N.

Sample 2

Input	Output
101	No

As $S(101) = 1 + 0 + 1 = 2$, $S(N)$ does not divide N.

Sample 3

Input	Output
999999999	Yes

Problem B. First Player

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

There are N people numbered $1, 2, \ldots, N$, sitting in this clockwise order around a round table. In particular, person 1 is sitting next to person N in the clockwise direction.

For each $i = 1, 2, \ldots, N$, person i has a name S_i and an age A_i. Here, no two people have the same name or the same age.

Starting from the youngest person, print the names of all N people in the order of their seating positions in clockwise order.

Constraints

- $2 \leq N \leq 100$
- N is an integer.
- S_i is a string of length between 1 and 10, consisting of lowercase English letters.
- $i \neq j \implies S_i \neq S_j$
- $0 \leq A_i \leq 10^9$
- A_i is an integer.
- $i \neq j \implies A_i \neq A_j$

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
S_1 \ \ A_1 \\
S_2 \ \ A_2 \\
\vdots \\
S_N \ \ A_N
\end{array}
$$

Output

Print N lines. For each $i = 1, 2, \ldots, N$, the i-th line should contain the name of the person sitting in the i-th position clockwise from the youngest person.

Sample 1

Input	Output
5 alice 31 bob 41 carol 5 dave 92 ellen 65	carol dave ellen alice bob

The youngest person is person 3. Therefore, starting from person 3, print the names in the clockwise order of their seating positions: person 3, person 4, person 5, person 1, and person 2.

Sample 2

Input	Output
2 takahashi 1000000000 aoki 999999999	aoki takahashi

Problem C. Maximize the Formula

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

You have decided to give an allowance to your child depending on the outcome of the game that he will play now.

The game is played as follows:

- There are three "integer panels", each with a digit between 1 and 9 (inclusive) printed on it, and one "operator panel" with a `+` printed on it.
- The player should construct a formula of the form $X + Y$, by arranging the four panels from left to right. (The operator panel should not be placed at either end of the formula.)
- Then, the amount of the allowance will be equal to the resulting value of the formula.

Given the values A, B and C printed on the integer panels used in the game, find the maximum possible amount of the allowance.

Constraints

- All values in input are integers.
- $1 \leq A, B, C \leq 9$

Input

Input is given from Standard Input in the following format:

```
A B C
```

Output

Print the maximum possible amount of the allowance.

Sample 1

Input	Output
1 5 2	53

The amount of the allowance will be 53 when the panels are arranged as `52+1`, and this is the maximum possible amount.

Sample 2

Input	Output
9 9 9	108

Sample 3

Input	Output
6 6 7	82

Problem D. A to Z String

Time Limit 2000 ms
Mem Limit 262144 kB

Problem Statement

Snuke has decided to construct a string that starts with `A` and ends with `Z`, by taking out a substring of a string s (that is, a consecutive part of s).

Find the greatest length of the string Snuke can construct. Here, the test set guarantees that there always exists a substring of s that starts with `A` and ends with `Z`.

Constraints

- $1 \leqq |s| \leqq 200{,}000$
- s consists of uppercase English letters.
- There exists a substring of s that starts with `A` and ends with `Z`.

Input

The input is given from Standard Input in the following format:

```
s
```

Output

Print the answer.

Sample 1

Input	Output
QWERTYASDFZXCV	5

By taking out the seventh through eleventh characters, it is possible to construct `ASDFZ`, which starts with `A` and ends with `Z`.

Sample 2

Input	Output
ZABCZ	4

Sample 3

Input	Output
HASFJGHOGAKZZFEGA	12

Problem E. Caesar Cipher

Time Limit 2000 ms

Problem Statement

Takahashi has a string S consisting of lowercase English letters.

On this string, he will do the operation below just once.

- First, choose a non-negative integer K.
- Then, shift each character of S to the right by K (see below).

Here,

- `a` shifted to the right by 1 is `b`;
- `b` shifted to the right by 1 is `c`;
- `c` shifted to the right by 1 is `d`;
- $\cdots$
- `y` shifted to the right by 1 is `z`;
- **`z` shifted to the right by 1 is `a`.**

For example, `b` shifted to the right by 4 is `f`, and `y` shifted to the right by 3 is `b`.

You are given a string T. Determine whether Takahashi can make S equal T by the operation above.

Constraints

- Each of S and T is a string of length between 1 and 10^5 (inclusive) consisting of lowercase English letters.
- The lengths of S and T are equal.

Input

Input is given from Standard Input in the following format:

```
S
T
```

Output

If Takahashi can make S equal T, print `Yes`; if not, print `No`.

Sample 1

Input	Output
abc ijk	Yes

When Takahashi chooses $K = 8$,

- `a` is shifted to the right by 8 and becomes `i`,
- `b` is shifted to the right by 8 and becomes `j`,
- `c` is shifted to the right by 8 and becomes `k`,

and now S and T are equal.
Therefore, he can make S equal T, so `Yes` should be printed.

Sample 2

Input	Output
z a	Yes

Choosing $K = 1$ makes S and T equal.
Note that the letter on the right of `z` is `a`.

Sample 3

Input	Output
ppq qqp	No

There is no non-negative integer K that he can choose to make S equal T, so `No` should be printed.

Sample 4

Input	Output
atcoder atcoder	Yes

Choosing $K = 0$ makes S and T equal.

Problem F. Pizza

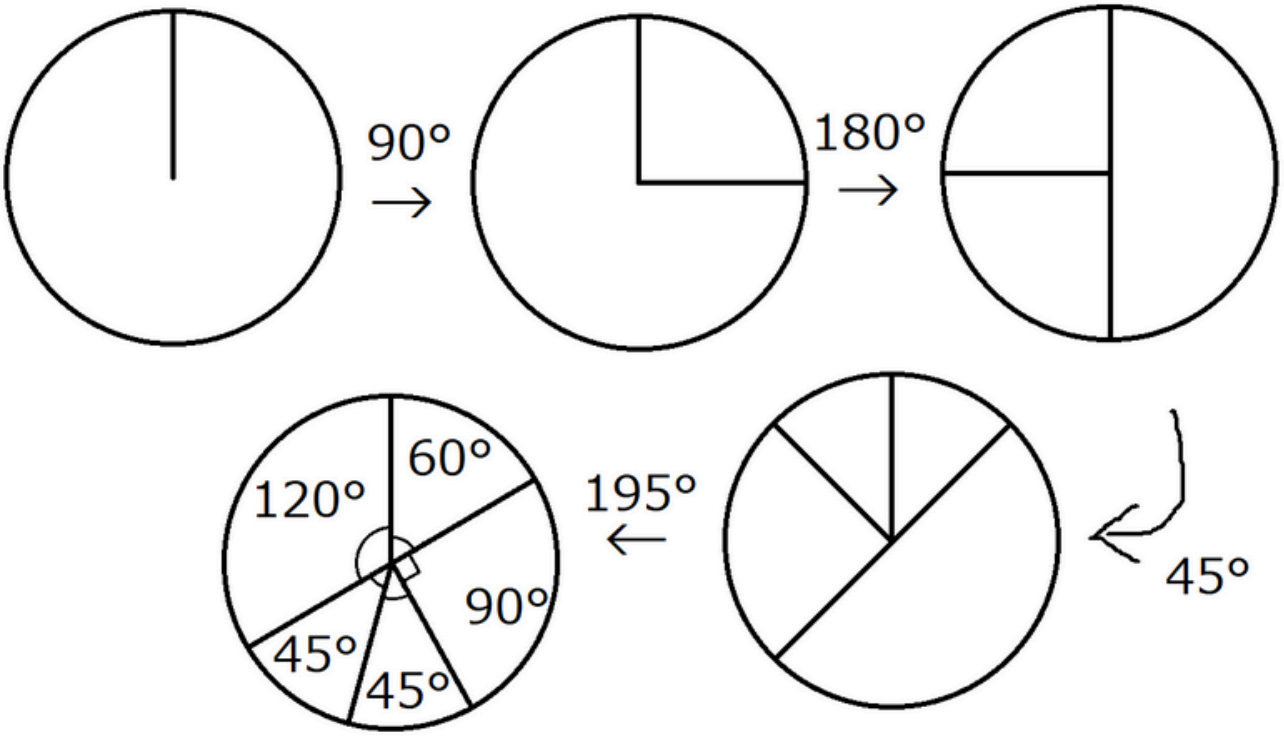
Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

We have a circular pizza.
Takahashi will cut this pizza using a sequence A of length N, according to the following procedure.

- First, make a cut from the center in the 12 o'clock direction.
- Next, do N operations. The i-th operation is as follows.
  - Rotate the pizza A_i degrees clockwise.
  - Then, make a cut from the center in the 12 o'clock direction.

For example, if $A = (90, 180, 45, 195)$, the procedure cuts the pizza as follows.



Find the center angle of the largest pizza after the procedure.

Constraints

- All values in input are integers.
- $1 \leq N \leq 359$

- $1 \leq A_i \leq 359$
- There will be no multiple cuts at the same position.

Input

Input is given from Standard Input in the following format:

$$N$$
$$A_1 \ A_2 \ \ldots \ A_N$$

Output

Print the answer as an integer.

Sample 1

Input	Output
4 90 180 45 195	120

This input coincides with the example in the Problem Statement.
The center angle of the largest pizza is 120 degrees.

Sample 2

Input	Output
1 1	359

Sample 3

Input	Output
10 215 137 320 339 341 41 44 18 241 149	170

Problem G. Max MEX

Time Limit 2000 ms

Problem Statement

You are given a length-N sequence of non-negative integers.
When B is a sequence obtained by choosing K elements from A and concatenating them without changing the order, find the maximum possible value of $MEX(B)$.

Here, for a sequence X, we define $MEX(X)$ as the unique non-negative integer m that satisfies the following conditions:

- Every integer i such that $0 \leq i < m$ is contained in X.
- m is not contained in X.

Constraints

- All values in the input are integers.
- $1 \leq K \leq N \leq 3 \times 10^5$
- $0 \leq A_i \leq 10^9$

Input

The input is given from Standard Input in the following format:

N K
A_1 A_2 $\ldots$ A_N

Output

Print the answer.

Sample 1

Input	Output
7 3 2 0 2 3 2 1 9	3

In this input, $A = (2, 0, 2, 3, 2, 1, 9)$, and you obtain the sequence B by picking $K = 3$ elements. For example,

- If the 1-st, 2-nd, and 3-rd elements are chosen, $MEX(B) = MEX(2, 0, 2) = 1$.
- If the 3-rd, 4-th, and 6-th elements are chosen, $MEX(B) = MEX(2, 3, 1) = 0$.
- If the 2-nd, 6-th, and 7-th elements are chosen, $MEX(B) = MEX(0, 1, 9) = 2$.
- If the 2-nd, 3-rd, and 6-th elements are chosen, $MEX(B) = MEX(0, 2, 1) = 3$.

The maximum possible MEX is 3.

Problem H. Perfect Bus

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

A bus is in operation. The number of passengers on the bus is always a non-negative integer.

At some point in time, the bus had zero or more passengers, and it has stopped N times since then. At the i-th stop, the number of passengers increased by A_i. Here, A_i can be negative, meaning the number of passengers decreased by $-A_i$. Also, no passengers got on or off the bus other than at the stops.

Find the minimum possible current number of passengers on the bus that is consistent with the given information.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $-10^9 \leq A_i \leq 10^9$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$N$$
$$A_1 \ A_2 \ \ldots \ A_N$$

Output

Print the answer.

Sample 1

Input	Output
4 3 -5 7 -4	3

If the initial number of passengers was 2, the current number of passengers would be $2 + 3 + (-5) + 7 + (-4) = 3$, and the number of passengers on the bus would have always been a non-negative integer.

Sample 2

Input	Output
5 0 0 0 0 0	0

Sample 3

Input	Output
4 -1 1000000000 1000000000 1000000000	3000000000

Problem I. Five Transportations

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

In 2028 and after a continuous growth, AtCoder Inc. finally built an empire with six cities (City $1, 2, 3, 4, 5, 6$)!

There are five means of transport in this empire:

- Train: travels from City 1 to 2 in one minute. A train can occupy at most A people.
- Bus: travels from City 2 to 3 in one minute. A bus can occupy at most B people.
- Taxi: travels from City 3 to 4 in one minute. A taxi can occupy at most C people.
- Airplane: travels from City 4 to 5 in one minute. An airplane can occupy at most D people.
- Ship: travels from City 5 to 6 in one minute. A ship can occupy at most E people.

For each of them, one vehicle leaves the city at each integer time (time $0, 1, 2, \ldots$).

There is a group of N people at City 1, and they all want to go to City 6.
At least how long does it take for all of them to reach there? You can ignore the time needed to transfer.

Constraints

- $1 \leq N, A, B, C, D, E \leq 10^{15}$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

N
A
B
C

D
E

Output

Print the minimum time required for all of the people to reach City 6, in minutes.

Sample 1

Input	Output
5 3 2 4 3 5	7

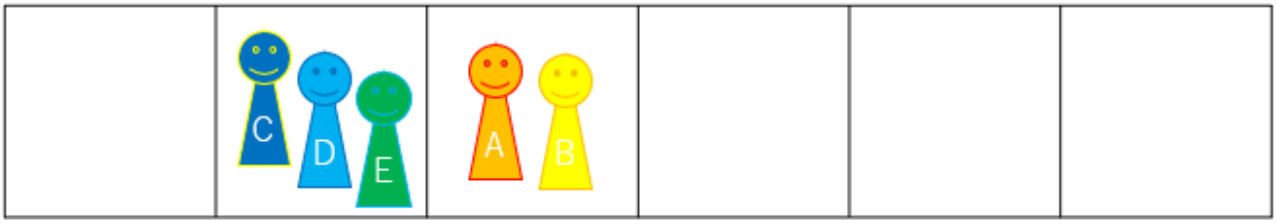
One possible way to travel is as follows. First, there are $N = 5$ people at City 1, as shown in the following image:

TIME: 0

In the first minute, three people travels from City 1 to City 2 by train. Note that a train can only occupy at most three people.

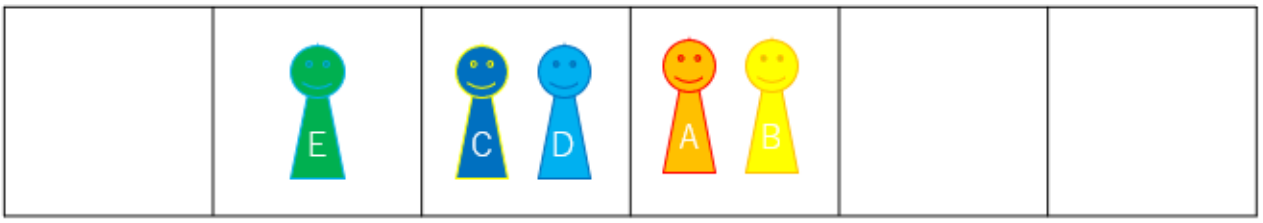
TIME: 1

In the second minute, the remaining two people travels from City 1 to City 2 by train, and two of the three people who were already at City 2 travels to City 3 by bus. Note that a bus can only occupy at most two people.



TIME: 2

In the third minute, two people travels from City 2 to City 3 by train, and another two people travels from City 3 to City 4 by taxi.



TIME: 3

From then on, if they continue traveling without stopping until they reach City 6, all of them can reach there in seven minutes.
There is no way for them to reach City 6 in 6 minutes or less.

Sample 2

Input	Output
10 123 123 123 123 123	5

All kinds of vehicles can occupy $N = 10$ people at a time. Thus, if they continue traveling without stopping until they reach City 6, all of them can reach there in five minutes.

Sample 3

Input	Output
10000000007 2 3 5 7 11	5000000008

Note that the input or output may not fit into a 32-bit integer type.

Problem J. Prison

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

We have N ID cards, and there are M gates.

We can pass the i-th gate if we have one of the following ID cards: the L_i-th, $(L_i + 1)$-th, ..., and R_i-th ID cards.

How many of the ID cards allow us to pass all the gates alone?

Constraints

- All values in input are integers.
- $1 \leq N \leq 10^5$
- $1 \leq M \leq 10^5$
- $1 \leq L_i \leq R_i \leq N$

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ \ M \\
L_1 \ \ R_1 \\
L_2 \ \ R_2 \\
\vdots \\
L_M \ \ R_M
\end{array}
$$

Output

Print the number of ID cards that allow us to pass all the gates alone.

Sample 1

Input	Output
4 2 1 3 2 4	2

Two ID cards allow us to pass all the gates alone, as follows:

- The first ID card does not allow us to pass the second gate.
- The second ID card allows us to pass all the gates.
- The third ID card allows us to pass all the gates.
- The fourth ID card does not allow us to pass the first gate.

Sample 2

Input	Output
10 3 3 6 5 7 6 9	1

Sample 3

Input	Output
100000 1 1 100000	100000